1(i1): Player A's mixed safety strategy $(1-\bar{\lambda}, \bar{\lambda})$ minimises

$$\lambda \to (3(1-\lambda)+\lambda) \wedge ((1-\lambda)+4\lambda)$$

over λ, $0 \leq \lambda \leq 1$. From the graph, the minimising $\bar{\lambda}$ occurs at the intersection point, i.e., when $3(1-\bar{\lambda})+\bar{\lambda} = (1-\bar{\lambda})+4\bar{\lambda}$.

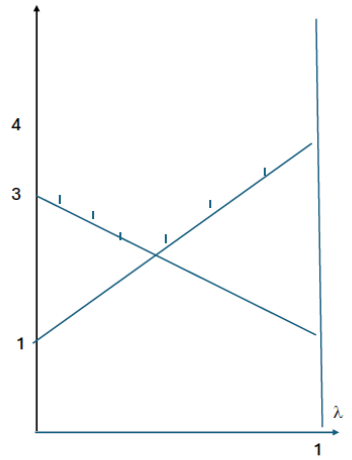
This gives $\bar{\lambda} = \frac{2}{5}$. A's safety strategy is therefore $(\frac{3}{5}, \frac{2}{5})$.

Player B's mixed safety strategy $(1-\bar{\mu}, \bar{\mu})$ minimises

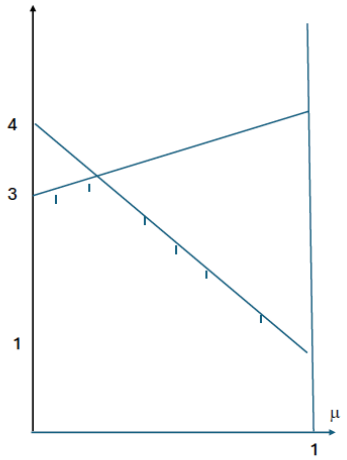
$$\mu \to (3(1-\mu)+4\mu) \wedge (4(1-\mu)+\mu)$$

over μ, $0 \leq \mu \leq 1$. From the graph, the maximising $\bar{\mu}$ occurs at the intersection point, i.e., when $3(1-\bar{\mu})+4\bar{\mu} = 4(1-\bar{\mu})+\bar{\mu}$.

This gives $\bar{\mu} = \frac{1}{4}$. B's safety strategy is therefore $(\frac{3}{4}, \frac{1}{4})$.



P_1's safety strategy



P_2's safety strategy

For $((1-\bar{\lambda}, \bar{\lambda}), (1-\bar{\mu}), \bar{\mu}))$ to be a Nash equilibrium, we require, in particular, that the function

$$\lambda \to [(1-\lambda, \lambda)] \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} \frac{3}{4} \\ \frac{1}{4} \end{bmatrix} = [(1-\lambda, \lambda)] \begin{bmatrix} \frac{10}{4} \\ \frac{7}{4} \end{bmatrix} = \frac{10}{4} - \frac{3}{4}\lambda$$

achieves a minimum over $[0,1]$ at $\lambda = \frac{2}{5}$. This is clearly not the case and so $((1-\bar{\lambda}, \bar{\lambda}), (1-\bar{\mu}, \bar{\mu}))$ is not a Nash equilibrium.

1(i2):

- For $((1-\lambda', \lambda'), (1-\mu', \mu'))$ to be an 'interior' Nash equilibrium, in mixed strategies, we require that

$$(d/d\lambda) r(\lambda') = 0 \text{ and } (d/d\mu) s(\lambda') = 0,$$

where $r(\lambda)$ and $s(\mu)$ are the functions:

$$r(\lambda) = [(1-\lambda, \lambda)] \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1-\mu' \\ \mu' \end{bmatrix} \text{ and } s(\mu) = [(1-\lambda', \lambda')] \begin{bmatrix} 3 & 4 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} 1-\mu \\ \mu \end{bmatrix}$$

This gives

$$[-1, +1] \begin{bmatrix} 3 & 1 \\ 1 & 4 \end{bmatrix} \begin{bmatrix} 1-\mu' \\ \mu' \end{bmatrix} = [(1-\lambda', \lambda')] \begin{bmatrix} 3 & 4 \\ 4 & 1 \end{bmatrix} \begin{bmatrix} -1 \\ +1 \end{bmatrix} = 0.$$

Hence $-2\mu' + 3(1-\mu') = 0$ and $(1-\lambda') - 3\lambda' = 0$.

So the Nash equilibrium $((1-\lambda', \lambda'), (1-\mu', \mu') = ((\frac{3}{4}, \frac{1}{4}), (\frac{3}{5}, \frac{2}{5}))$

- For zero sum matrix 2-player matrix games in mixed strategies, or games that can be converted to such games, the Nash equilibria coincide with pairs of safety strategies. But for the above choice of A and B, the game is not zero sum, and we can not expect the strategies to be the same.

1(ii): Take $\bar{x}$ to be a safety strategy for P_1. Define $\bar{p} := \max_{y \in \Lambda^{(m)}} \bar{x}^T A y$. By considering $y = [1, 0, \ldots, 0]^T$, $y = [0, 1, \ldots, 0]^T$, ..., $y = [0, 0, \ldots, 1]^T$, respectively, we can deduce that

$$\bar{p} \geq (\bar{x}^T A)_i, \text{ for all } i \implies p\, 1^{(n)} \geq \bar{x}^T A\,.$$

Since $\bar{x} \in \Lambda^{(n)}$, we have shown that $(\bar{p}, \bar{x})$ satisfy the constraints of (L). By definition of 'safety strategy',

$$\bar{p} = \max_{y \in \Lambda} \bar{x}^T A y \leq \max_{y \in \Lambda} x^T A y \text{ for all } x \in \Lambda^{(n)}.$$

Now take any $(p, x) \in R \times \Lambda^{(n)}$ satisfying the constraints of (L). Then for any $y \in \Lambda^{(m)}$,

$$p = p\, 1^{(n)} y \geq x^T A y\,.$$

Hence

$$p \geq \max_{y \in \Lambda^{(m)}} x^T A y \geq \bar{p}.$$

This means $(\bar{p}, \bar{x})$ is a minimiser for (L).

2: Write L :='left' and R =: 'right'. To describe A's strategy space, we write

$A(xyz)$:= 'In stage 1 A chooses x, and in stage 2 A chooses y if B chooses L and z if B chooses R in stage 1.'

A's strategy space has 8 elements. They are

$$A(LLL),\ A(LRL),\ A(LLR),\ A(LRR),\ A(RLL),\ A(RRL),\ A(RLR) \text{ and } A(RRR)$$

To describe B strategy space, we write

$B(x'y')$:= 'In stage 1 B chooses x', and in stage 2 B chooses y'.'

B strategy space has 4 elements. They are

$$B(LL),\ B(LR),\ B(RL),\ \text{and } B(RR).$$

The following table shows path down the tree associated with every choice of policy pair $A(xyz)$,

$B\backslash A$	A(LLL)	A(LRL)	A(LLR)	A(LRR)	A(RLL)	A(RRL)	A(RLR)	A(RRR)
B(LL)	LLLL	LLRL	LLLL	LLRL	RLLL	RLRL	RLLL	RLRL
B(RL)	LRLL	LRLL	LRRL	LRRL	RRLL	RRLL	RRRL	RRRL
B(LR)	LLLR	LLRR	LLLR	LLRR	RLLR	RLRR	RLLR	RLRR
B(RR)	LRLR	LRLR	LRRR	LRRR	RRLR	RRLR	RRRR	RRRR

Now read out the pay-off's from the diagram

$B\backslash A$	A(LLL)	A(LRL)	A(LLR)	A(LRR)	A(RLL)	A(RRL)	A(RLR)	A(RRR)
B(LL)	2	3	2	3	3	4	3	4
B(RL)	3	3	1	1	2	2	3	3
B(LR)	1	4	1	4	2	5	2	5
B(RR)	4	4	0	0	1	1	-1	-1

The table shows that there is one Nash equilibrium which is $(A(LRL), B(RL))$. This means the sequence of actions is: A chooses L and B chooses R in stage 1. Then A chooses L and B chooses L is stage two.

The pay-off is 3.

We see that the row of $A(LRL)$ is not dominated by any other row, and the column of $B(RL)$ is not dominated by any other column. It follows that equilibrium is admissible.

3(i): The pay-off functions are

$$J_1(p_1,p_2) = p_1 \times (4 - 2p_1 - p_2) - \tfrac{1}{2}(4 - 2p_1 - p_2)^2,$$
$$J_2(p_1,p_2) = p_2 \times (4 - 2p_2 - p_1) - \tfrac{1}{4}(4 - 2p_2 - p_1)^2.$$

3(ii): The partial derivatives are

$$(\partial J_1/\partial p_1)(p_1,p_2) = 4 - 2p_1 - p_2 - 2p_1 + 2(4 - 2p_1 - p_2)$$
$$(\partial J_2/\partial p_2)(p_1,p_2) = 4 - 2p_2 - p_1 - 2p_2 + (4 - 2p_2 - p_1)$$

$$\text{These vanish when } 12 - 8p_1 - 3p_2 = 0 \text{ and } 8 - 6p_2 - 2p_1 = 0 \tag{1}$$

The Nash equilibrium $(\bar{p}_1, \bar{p}_2)$ is the solution to these equations:

$$(\bar{p}_1, \bar{p}_2) = (\frac{8}{7}, \frac{20}{21}).$$

3(iii): From (1), the best strategies for C_1 when C_2 chooses p_2 and for C_2 when C_1 chooses p_1 are

$$p_1(p_2) = \frac{12 - 3p_2}{8} \quad \text{and} \quad p_2(p_1) = \frac{4 - p_1}{3}.$$

The graphs of these functions are the reaction curves of C_1 and C_2. (See Figure 1). The intersection $(\bar{p}_1, \bar{p}_2) = (\frac{8}{7}, \frac{20}{21})$ is indeed the Nash equilibrium.

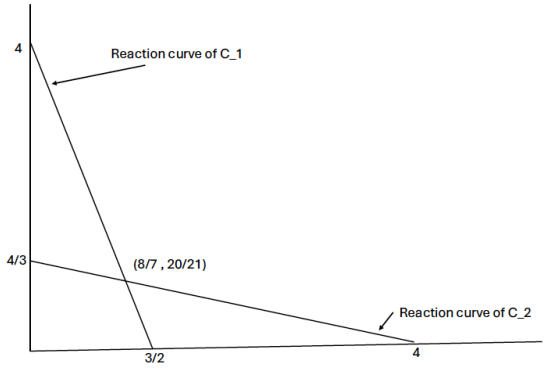


Figure 1: Plot of reaction curves.

3(iv): If C_2 chooses optimal price (given C_1 chooses p_1), the pay-off for C_1 is

$$\begin{aligned} J_1(p_1, p_2(p_1)) &= p_1(4 - 2p_1 - \tfrac{4}{3} - \tfrac{1}{3}p_1) - \tfrac{1}{2}(4 - 2p_1 - \tfrac{4}{3} - \tfrac{1}{3}p_1)^2 \\ &= p_1(\tfrac{8}{3} - \tfrac{7}{3}p_1) - \tfrac{1}{2}(\tfrac{8}{3} - \tfrac{7}{3}p_1)^2. \end{aligned}$$

Setting the gradient of $p_1 \to J_1(p_1, p_2(p_1))$ to zero gives

$$\begin{aligned} (d/dp_1)J_1(p_1, p_2(p_1)) &= (\tfrac{8}{3} - \tfrac{7}{3}p_1) - \tfrac{7}{3}p_1 + \tfrac{7}{3}(\tfrac{8}{3} - \tfrac{7}{3}p_1) \\ &= (\tfrac{8}{3} + \tfrac{56}{9}) - (\tfrac{7}{3} + \tfrac{7}{3} + \tfrac{7}{3} \times \tfrac{7}{3})p_1 = 0. \end{aligned}$$

This gives $p_1 = \frac{80}{91}$. This is the leader's optimal Stackelberg strategy.

The follower's response is $p_2 = \frac{4 - p_1}{3} = \frac{4 - \frac{80}{91}}{3} = \frac{284}{273}$.

4(i): Write $a(x) = \text{col}\{a_1, \ldots, a_n\}$. Then $\bar{x} \in \Lambda^{(n)}$ is a Nash equilibrium when

$$\bar{x}^T a(\bar{x}) \geq x^T a(\bar{x}), \quad \text{for all } x \in \Lambda^{(n)}.$$

The Nash equilibrium is stable means that, for some $\epsilon > 0$,

$$\bar{x}^T a(x) > x^T a(x), \text{ for all } x \in \Lambda^{(n)} \text{ s.t. } x \neq \bar{x} \text{ and } ||x - \bar{x}|| \leq \epsilon.$$

4(ii): Take any $x \in \Lambda^{(n)}$, $x \neq \bar{x}$. Then, since $x \neq \bar{x}$ and $\bar{x}$ is a Nash equilibrium,

$$\begin{aligned} \bar{x}^T A x - x^T A x &= (\bar{x} - x)^T A \bar{x} - (\bar{x} - x)^T A (\bar{x} - x) \\ &\geq 0 - (\bar{x} - x)^T A (\bar{x} - x) > 0. \end{aligned}$$

This strict inequality confirms that $\bar{x}$ is stable.

4(iii): Notice that, in the specifications of the fitness functions, we can replace 10 by 0, since the Nash equilibria are unaffected by adding the same constant to all the fitness functions. The game then is a matrix game with fitness matrix

$$A = -\begin{bmatrix} 1 & k_0 & k_0 & k_0 \\ k_1 & 1 & 0 & 0 \\ k_1 & 0 & 1 & 0 \\ k_1 & 0 & 0 & 1 \end{bmatrix}.$$

Take $\bar{x} = [\bar{x}_1, \ldots, \bar{x}_4]$ to be an interior Nash equlibrium. Following the hint, we assume $\bar{x}_2 = \bar{x}_3 = \bar{x}_4$. Using the interior equilibrium criterion 'all fitness functions must take the same value' and the fact that $\bar{x} \in \Lambda^{(4)}$ we deduce that

$$\bar{x}_1 + 3k_0\bar{x}_2 = \bar{x}_2 + k_1\bar{x}_1 \text{ and } \bar{x}_1 + 3\bar{x}_2 = 1\,.$$

We see that $\bar{x}_1 = (1 - 3\bar{x}_2)$. Eliminating $\bar{x}_1$ from the equations then gives

$$\bar{x}_2 = \frac{1 - k_1}{3(1 - k_1) + (1 - 3k_0)} \text{ and hence } \bar{x}_1 = \frac{1 - 3k_0}{3(1 - k_1) + (1 - 3k_0)}$$

Setting $k_0 = 1/12$ and $k_1 = 3/4$ and using the fact that $\bar{x}_2 = \bar{x}_3 = \bar{x}_4$, we arrive at

$$(\bar{x}_1, \bar{x}_2, \bar{x}_3, \bar{x}_4) = (1/2, 1/6, 1/6, 1/6).$$

By part (ii), this equilibrium will be stable if A is a negative definite matrix, i.e.

$$y^T A y < 0, \text{ for all } y \neq 0\,.$$

5(i): Given a state trajectory $\{x_t\}$, let $z_t = [x_t, a]$ for $t = 1, \ldots, N$. Then $\{z_t\}$ solves the state equation

$$z_{t+1} = Az_t + bu_t$$

with initial condition $z_0 = (0, a)$ when $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$.

Also, $z_N^T G z_N = [x_N, a] G \begin{bmatrix} x_N \\ 0 \end{bmatrix} = x_N^2$ when $G = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$. It follows $\{\bar{u}_t\}$ is a minimiser for (R), with the above choices of A, b and G.

From the given formulae for the solution to the (LQ) problem the optimal $\{\bar{u}_t\}$ and $\{\bar{z}_t = (\bar{x}_t, a)\}$ are given by the feedback relation

$$\begin{aligned} \bar{u}_{t-1} &= -(I + b^T P_t b)^{-1} b^T P_t A[\bar{x}_{t-1}, a]^T \\ &= -(1 + p_t^{11})^{-1} [1 \quad 0] \begin{bmatrix} p_t^{11} & p_t^{12} \\ p_t^{12} & p_t^{22} \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \bar{x}_{t-1} \\ a \end{bmatrix} \\ &= -\tfrac{1}{1+p_t^{11}} \times (p_t^{11} x_{t-1} + (p^{11} + p_t^{12}) a) \end{aligned}$$

in which $P_t = \begin{bmatrix} p_t^{11} & p_t^{12} \\ p_t^{12} & p_t^{22} \end{bmatrix}$ satisfies the recursion: for $t = N, N-1, \ldots, 1$

$$\begin{aligned} P_{t-1} &= A^T \Big(P_t - P_t b (I + b^T P_t b)^{-1} b^T P_t \Big) A \\ &= \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \left(P_t - (1 + p_t^{11})^{-1} \begin{bmatrix} (p_t^{11})^2 & p_t^{11} p_t^{12} \\ p_t^{11} p_t^{12} & (p_t^{12})^2 \end{bmatrix} \right) \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \end{aligned} \tag{2}$$

with starting condition $P_N = G = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

The minimum cost is $([0, a] P_0 \begin{bmatrix} 0 \\ a \end{bmatrix} = p_0^{11} \times a^2$.

5(ii): If the leader chooses a, the follower will choose the solution to problem (P) (with parameter a) and the follower's minimum cost will be $p_0^{11} a^2$. Here, p_0^{11} is obtained by solving the recursion (2). The leader's cost will then be

$$p_0^{11} \times a^2 - 3a\,.$$

This is minimised when $a = \frac{3}{2p_0^{11}}$, which is the Stackelberg strategy for the leader.